

Finite generation of canonical rings II

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Outline of the talk

1 Introduction

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- 2 The MMP with scaling

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- 3 The structure of the proof

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Notation

- $X \subset \mathbb{P}_{\mathbb{C}}^N$ is defined by homogeneous polynomials $P_1, \dots, P_t \in \mathbb{C}[x_0, \dots, x_N]$.
- X, X' are birational if they have isomorphic open subsets $U \cong U'$.
- If X is smooth of dimension n , then T_X denotes the **tangent bundle** of X and $\omega_X = \Lambda^n T_X^\vee$ is the **canonical line bundle**.
- $H^0(X, \omega_X^{\otimes m})$ denotes the space of global pluricanonical sections (locally given by $f(z)(dz_1 \wedge \dots \wedge dz_n)^{\otimes m}$).
- K_X denotes (a choice of) the canonical divisor i.e. a formal linear combination of codimension 1 subvarieties of X given by the zeroes of a meromorphic section $s \neq 0$ of ω_X .

The canonical ring

- If X and X' are birational, then $\forall m \geq 0$

$$H^0(X, \omega_X^{\otimes m}) \cong H^0(X', \omega_{X'}^{\otimes m}).$$

- The **canonical ring** of X is given by

$$R(K_X) = \bigoplus_{m \in \mathbb{N}} H^0(X, \omega_X^{\otimes m}).$$

- One would like to understand the structure of this ring and to use its features to classify complex projective varieties.

Finite generation

Today I will illustrate the techniques involved in the proof of the following result

Theorem (Birkar-Cascini-Hacon-M^cKernan / Siu)

Let X be a smooth projective variety, then $R(K_X)$ is finitely generated.

The two proofs are independent. Our proof is algebraic and uses the ideas of the MMP (minimal model program). Siu's proof is analytic and requires X to be of general type (i.e. K_X is big).

KLT Log Pairs

- We consider log pairs (X, Δ) where X is a normal $\mathbb{Q}$ -factorial variety and $\Delta = \sum \delta_i \Delta_i$ is a divisor with $\delta_i \in \mathbb{R}$ and $0 \leq \delta_i < 1$.
- Then $K_X + \Delta$ is $\mathbb{R}$ -Cartier and so $(K_X + \Delta) \cdot C$ and $f^*(K_X + \Delta)$ make sense.
- Consider a map $f : X' \rightarrow X$ such that X' is smooth and $f^*\Delta + \text{Exc}(f)$ has simple normal crossings support. Write

$$K_{X'} + \Delta' = f^*(K_X + \Delta).$$

- (X, Δ) is KLT if each coefficient of Δ' is less than 1.
- A KLT pair (X, Δ) is a minimal model if $K_X + \Delta$ is nef (i.e. $(K_X + \Delta) \cdot C \geq 0$ for any curve $C \subset X$).

Notation II

- A divisor D is big if $D \sim_{\mathbb{R}} A + B$ where A is ample and $B \geq 0$.
- A divisor D is pseudo-effective if it is a limit of big divisors.
- Note that any nef divisor is pseudo-effective.
- A $\mathbb{Q}$ -divisor D is semiample if for some $m > 0$ the divisor mD is base point free.
- If D is semiample, then $R(D) = \bigoplus_{i \geq 0} H^0(\mathcal{O}_X(iD))$ is finitely generated.
- Any semiample divisor is nef but the converse is not true.
- By the Base Point Free Theorem, if $K_X + \Delta$ is KLT and Δ is big, then $K_X + \Delta$ is semiample.

A Theorem of Fujino-Mori

Recall that Mori and Fujino have shown:

Theorem

Let (X, Δ) be any KLT $\mathbb{Q}$ -factorial pair with $\Delta \in \text{Div}_{\mathbb{Q}}(X)$. Then there exists a **big** KLT $\mathbb{Q}$ -factorial pair (Y, Γ) with $\Gamma \in \text{Div}_{\mathbb{Q}}(Y)$ such that

$$R(K_X + \Delta)^{(m)} \cong R(K_Y + \Gamma)^{(m)}$$

for any sufficiently divisible integer $m > 0$.

It follows that to show that $R(K_X)$ is finitely generated, it suffices to prove the corresponding statement for big KLT log pairs.

The main Theorem

Theorem (Birkar-Cascini-Hacon-McKernan)

Let (X, Δ) be any KLT $\mathbb{Q}$ -factorial pair with $\Delta \in \text{Div}_{\mathbb{R}}(X)$. If Δ is big and $K_X + \Delta$ is pseudo-effective, then (X, Δ) has a minimal model.

More precisely, there is a sequence of well understood operations (flips and divisorial contractions) $X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_n$ such that $R(K_X + \Delta) \cong R(K_{X_n} + \Delta_n)$ and $K_{X_n} + \Delta_n$ is nef and KLT. By the base-point-free theorem, $K_{X_n} + \Delta_n$ is semiample and so $R(K_X + \Delta)$ is finitely generated.

- Note that our theorem also applies when $K_X + \Delta$ is big and KLT.
- Since $K_X + \Delta$ is big, we have $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$.
- Therefore for any rational number $0 < \epsilon \ll 1$, we have

$$K_X + \Delta' = K_X + \Delta + \epsilon D \sim_{\mathbb{R}} (1 + \epsilon)(K_X + \Delta)$$

where (X, Δ') is KLT and Δ' is big.

- It follows that if $K_X + \Delta$ is a big and KLT $\mathbb{Q}$ -divisor, then $R(K_X + \Delta)$ is finitely generated.
- By the Theorem of Mori-Fujino, if X is a smooth projective variety, then $R(K_X)$ is finitely generated.

Outline of the talk

- 1 Introduction
- 2 The MMP with scaling
- 3 The structure of the proof

The MMP with scaling

- We begin with a $\mathbb{Q}$ -factorial KLT pair (X, Δ) such that Δ is big and $K_X + \Delta$ is pseudo-effective. We allow $\Delta \in \text{Div}_{\mathbb{R}}(X)$.
- These conditions will be preserved by the birational maps that we will consider.
- We wish to find a birational model $X \dashrightarrow X'$ such that $K_{X'} + \Delta'$ is nef.
- Assume that A is ample and $K_X + \Delta + A$ is nef.
- Let $\lambda = \inf\{t \geq 0 \mid K_X + \Delta + tA\} \text{ is nef.}$
- If $\lambda = 0$ we STOP ($K_X + \Delta$ is nef).

The MMP with scaling II

- Otherwise, by the cone theorem, there is an extremal ray R such that $(K_X + \Delta + \lambda A) \cdot R = 0$ and $(K_X + \Delta + tA) \cdot R < 0$ for $0 \leq t < \lambda$.
- Let $\text{cont}_R : X \rightarrow Z$ be the corresponding morphism which contracts a curve C iff $[C] = R$.
- If cont_R is not birational, we STOP. (We have a Mori-Fano fiber space and $R(K_X + \Delta) \cong \mathbb{C}$. This only happens when $K_X + \Delta$ is not pseudo-effective.)
- If cont_R is birational and divisorial (i.e. it contracts a divisor), then we replace X by the image of the corresponding contraction and we may continue this procedure.

The MMP with scaling III

- If cont_R is birational and small, (i.e. $\text{Exc}(\text{cont}_R)$ has codimension ≥ 2) then the image of the corresponding contraction is not KLT. Therefore, we can NOT replace X by the corresponding contraction!
- Instead we replace (X, Δ) by the corresponding flip $\phi : X \dashrightarrow X^+$ (assuming it exists).
- Let $\Delta^+ = \phi_* \Delta$. The flip is a codimension 2 surgery which replaces $K_X + \Delta$ negative curves by $K_{X^+} + \Delta^+$ positive curves.

Flips

- More precisely, $f = \text{cont}_R : X \rightarrow Z$ is a flipping contraction i.e. a small birational morphism with $\rho(X/Z) = 1$ and $-(K_X + \Delta)$ ample over Z .
- The flip $f^+ : X^+ \rightarrow Z$ (if it exists) is a small birational morphism such that $\rho(X^+/Z) = 1$ and $K_{X^+} + \Delta^+$ is ample over Z .
- The flip $f^+ : X^+ \rightarrow Z$ (if it exists) is given by

$$X^+ = \text{Proj}_Z \bigoplus_{m \in \mathbb{N}} f_* \mathcal{O}_X([m(K_X + \Delta)]).$$

- Note that after perturbing Δ , we may assume that $\Delta \in \text{Div}_{\mathbb{Q}}(X)$.

Flips II

- The problem of existence of flips is local over Z and hence we may assume that Z is affine. It suffices then to show that $R(K_X + \Delta)$ is finitely generated. So this is a local version of the original problem.
- To run this MMP with scaling, we must therefore show that such flips exist and terminate (i.e. there is no infinite sequence of such flips).
- Note that after each divisorial contraction, the Picard number drops by 1 and so there are only finitely many divisorial contractions.

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- 1 Introduction
- 2 The MMP with scaling
- 3 The structure of the proof
 - Existence of minimal models
 - Non-vanishing
 - Finiteness of models

The structure of the proof

We proceed by induction on $n = \dim X$. We show that for an n -dimensional KLT pair (X, Δ) :

Claim 1. Flips exist.

Claim 2. If $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$ and Δ is big, then (X, Δ) has a minimal model.

Claim 3. If Δ is big and $K_X + \Delta$ is pseudo-effective, then there is an $\mathbb{R}$ -divisor D such that $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$.

Claim 4. The set of minimal models for KLT pairs with $A \leq \Delta \leq \Delta_0$ where A is ample and (X, Δ_0) is KLT, is finite.
All statements should be relative over a base.

The structure of the proof II

- Roughly speaking, the inductive structure of the proof is:

$$(1 - 4)_{n-1} \Rightarrow 1_n \Rightarrow 2_n \Rightarrow 3_n \Rightarrow 4_n.$$

- Note that that $(1 - 4)_{n-1} \Rightarrow 1_n$ was obtained by Hacon M^cKernan in 2005 using ideas of Shokurov and an extension result based on work of Siu, Kawamata and Tsuji. (This was discussed yesterday by James.)
- The remaining implications heavily use the ideas of the MMP established by Kawamata, Kollár, Mori, Reid, Shokurov and others.

Termination of MMP with scaling

- Since we are unable to show that an arbitrary sequence of flips must terminate, we run a MMP with scaling.
- The idea is that if A is sufficiently ample, then $K_X + \Delta + A$ is ample and so it is a minimal model. We then consider $K_X + \Delta + tA$ and let t go to 0.
- Proceeding this way we get a sequence of pairs $(X_i, \Delta_i + \lambda_i A_i)$ such that $K_{X_i} + \Delta_i + \lambda_i A_i$ is nef and trivial on some $K_{X_i} + \Delta_i$ extremal ray. We then flip/contract.
- Each $(X_i, \Delta_i + \lambda_i A_i)$ is a minimal model for $(X, \Delta + \lambda_i A)$.

Termination of MMP with scaling II

- If we could assume Claim 4_n , we would be done. (As Δ is big, we may assume that

$$A' \leq \Delta \leq \Delta + tA \leq \Delta + A$$

for some ample divisor A' .)

- Since we can only assume 4_{n-1} , we replace (X, Δ) by an appropriate DLT pair (X', Δ') and we run a MMP with scaling such that all flips/contractions are contained in $[\Delta']$.
- This step is similar to Shokurov's reduction to PL-flips.

Non-vanishing

- We must show that if Δ is big and $K_X + \Delta$ is pseudo-effective, then $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$.
- If $h^0(\mathcal{O}_X(m(K_X + \Delta) + A))$ is bounded (for A ample), then the result follows from work of Nakayama.
- If $h^0(\mathcal{O}_X(m(K_X + \Delta) + A))$ is un-bounded, then we can produce a log canonical center not contained in the stable base locus of $K_X + \Delta + \epsilon A$ for $0 < \epsilon \ll 1$.
- After replacing X by an appropriate birational model, we may then assume that (X, Δ) is PLT (“almost KLT”) and $S = [\Delta']$ is irreducible and not contained in the stable base locus of $K_X + \Delta + \epsilon A$ for any $0 < \epsilon \ll 1$.

Non-vanishing II

- Since $K_X + \Delta + \epsilon A$ is big, one sees that we may still run a $K_X + \Delta$ MMP with scaling of A .
- However, since we do not know that $K_X + \Delta$ is effective, this MMP might not terminate.
- By induction on the dimension, the flipping locus is eventually disjoint from S .
- Then $(K_X + \Delta + \epsilon A)|_S$ is nef for all $0 < \epsilon \ll 1$. Therefore, $K_S + \Delta_S = (K_X + \Delta)|_S$ is pseudo-effective, Δ_S is big and so $h^0(\mathcal{O}_S(m(K_S + \Delta_S))) > 0$ for some $m > 0$.
- Using Kawamata-Viehweg vanishing, we can lift this section to $m(K_X + \Delta)$.

Finiteness of models

Theorem

*Assume that (X, Δ_0) is KLT and A is an ample $\mathbb{Q}$ -divisor.
Then the set of isomorphism classes*

$$\{Y \mid Y \text{ is a min. model for } (X, \Delta); A \leq \Delta \leq \Delta_0\}$$

is finite.

The proof is based on ideas of Shokurov (1996). It requires a compactness argument and so it is necessary to work with $\mathbb{R}$ -divisors.

We will only show that there exist finitely many minimal models $\phi_i : X \dashrightarrow X_i$ such that for any $A \leq \Delta \leq \Delta_0$ there exists i such that X_i is a minimal model for (X, Δ) .

Finiteness of models II

- We pick $\Delta \in [A, \Delta_0]$ and work in a sufficiently small neighborhood $U = (\Delta_1, \Delta_2)$ of Δ .
- If $K_X + \Delta$ is not pseudo-effective, then (after shrinking U), $K_X + \Theta$ is not pseudo-effective for any $\Theta \in U$.
- Therefore, we may assume that $K_X + \Delta$ is pseudo-effective.
- By Claim 3, $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$ and so, by Claim 2, (X, Δ) has a minimal model.
- One sees that replacing (X, Δ) by its minimal model, we may assume that $K_X + \Delta$ is nef.
- By the base-point free theorem, there is a morphism $f : X \rightarrow Z$ such that $K_X + \Delta \sim_{\mathbb{R}} f^*H$ with H ample.

Finiteness of models III

- Working over Z , we may assume that $K_X + \Delta \sim_{\mathbb{R}, Z} 0$.
- Let $\Theta \in (\Delta, \Delta_i]$ sufficiently close to Δ . We may also assume that $K_X + \Theta$ is nef iff it is nef over Z . Therefore, if $\phi_i : X \dashrightarrow X_i$ is a minimal model for (X, Θ) over Z , then it is a minimal model for (X, Θ) (over $\text{Spec}(\mathbb{C})$).
- We may write

$$\Theta - \Delta = (K_X + \Theta) - (K_X + \Delta) \sim_{\mathbb{R}, Z} K_X + \Theta.$$

- If $K_X + \Theta$ is pseudo-effective, then it is pseudo-effective over Z and hence so is

$$K_X + \Delta_i \sim_{\mathbb{R}, Z} \Delta_i - \Delta = \lambda(\Theta - \Delta).$$

- Let $\phi_i : X \dashrightarrow X_i$ be a minimal model of (X, Δ_i) over Z . Then X_i is a minimal model of (X, Θ) .